AIM 340 and the Alman–Vassilevska Williams 3SUM algorithm

Kaiwen Jin
yui.ui.siki@gmail.com

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Abstract

Alman and Vassilevska Williams give a deterministic $O(N^{1.9992})$-time algorithm for integer 3SUM in the logarithmic-word RAM. We explain its application to the exact version in AIM Problem 340: given n distinct integers in $[-n^3, n^3]$, decide whether three pairwise distinct elements satisfy $a + b = c$. An elementary reduction uses $O(\log n)$ zero-sum instances of size at most $2n$, with a fixed polynomial bound on their entries. It gives total time $O(n^{2-1/2500})$ for every valid input. The fast algorithm is entirely due to Alman and Vassilevska Williams; this note records the correspondence with the stated target.

1 The target and the cited algorithm

We use Problem 340 of the AIM collection at the exact revision recorded in reference [2]. For each integer $n \geq 3$, its input is an explicitly listed set $S \subseteq [-n^3, n^3] \cap \mathbb{Z}$ with $|S| = n$. The answer is YES precisely when there are pairwise distinct $a, b, c \in S$ with $a + b = c$. The model is a classical word-RAM with $w = \Theta(\log n)$-bit words, large enough to store an input integer and an address. Usual word arithmetic, bit operations, comparisons and indexed memory accesses have unit cost. A single finite program must work at every size. All input-dependent preprocessing is counted.

For an admissible algorithm A, the target uses

$$T_A(n) = \max_{\substack{S \subseteq [-n^3, n^3] \cap \mathbb{Z} \\ |S| = n}} \mathbb{E}[\text{total word-RAM steps of } A(S)].$$

The answer must be correct with probability at least $2/3$ on each input. The conjecture excludes $T_A(n) = O(n^{2-\varepsilon})$ for every fixed $\varepsilon > 0$.

The primary result we use is Alman and Vassilevska Williams [1, Theorem 22, Section 3.4]. For every fixed constant ν, they give a deterministic algorithm for zero-sum 3SUM on N integers of absolute value at most N^ν, with total time $O(N^{1.9992})$. Their Theorem 2 specifies the standard word-RAM with $O(\log N)$-bit words. We fix $\nu = 6$ throughout. We denote this algorithm by Z. Its construction, preprocessing and running-time analysis belong to the cited work. The task here is to enforce distinctness of the original elements while using Z.

2 A reduction preserving distinct elements

First sort S. If $0 \in S$, test whether both x and $-x$ belong to S for some nonzero x. If so, return YES: $(x, -x, 0)$ is a valid distinct triple. Sorting and binary searches take $O(n \log n)$ total time.

Set

$$S^* = S \setminus \{0\}, \qquad m = |S^*|, \qquad U = n^3, \qquad Q = 4U + 1.$$

Assign the elements of S^* their sorted ranks $j = 0, \ldots, m-1$. For $t \geq 0$, write $\mathrm{bit}_t(j) = \lfloor j/2^t \rfloor \bmod 2$. Let $L = \lceil \log_2 m \rceil$; since $n \geq 3$, we have $m \geq 2$. For $0 \leq t < L$, form

$$\begin{aligned} T_t = &\{Q + x : x \in S^*,\ \mathrm{bit}_t(\mathrm{rank}(x)) = 0\} \\ &\cup \{2Q + x : x \in S^*,\ \mathrm{bit}_t(\mathrm{rank}(x)) = 1\} \\ &\cup \{-3Q - x : x \in S^*\}. \end{aligned} \tag{1}$$

Call Z on each T_t and return YES if any call does. If the initial stage and all calls return NO, return NO. The tags $1, 2, -3$ force a zero-sum triple to use one value from each of the three displayed sets. The rank bit then separates the two summands.

Lemma 1. *Every T_t has $N = 2m$ distinct integers, where $n \leq N \leq 2n$, and all its entries have absolute value at most N^6.*

Proof. The three sets in (1) lie respectively in

$$[3U + 1, 5U + 1], \qquad [7U + 2, 9U + 2], \qquad [-13U - 3, -11U - 3].$$

These intervals are disjoint, and each encoding within one set is injective. The first two sets together have m entries, and the last has m. Also $n - 1 \leq m \leq n$, whence $n \leq 2m \leq 2n$. Finally, for $n \geq 3$,

$$13n^3 + 3 \leq 14n^3 \leq n^6 \leq N^6.$$

Thus each call has the fixed polynomial integer bound required by Z. □

Proposition 2. *The reduction returns YES if and only if S contains pairwise distinct a, b, c with $a + b = c$.*

Proof. The initial opposite-pair test produces a valid witness. Suppose a call on T_t finds a zero-sum triple. Each term has the form $h_i Q + r_i$, with $h_i \in \{1, 2, -3\}$ and $|r_i| \leq U$. Since $|r_1 + r_2 + r_3| \leq 3U < Q$, zero sum forces $h_1 + h_2 + h_3 = 0$ and $r_1 + r_2 + r_3 = 0$.

Let k_1, k_2, k_3 count the tags $1, 2, -3$. Then $k_1 + k_2 + k_3 = 3$ and $k_1 + 2k_2 - 3k_3 = 0$. If $k_3 = 0$ the latter sum is positive; if $k_3 \geq 2$ it is negative. Hence $k_3 = 1$ and $k_1 = k_2 = 1$. The triple is therefore $(Q + a, 2Q + b, -3Q - c)$ for $a, b, c \in S^*$ with $a + b = c$. The rank-bit condition gives $a \neq b$. If $c = a$, then $b = 0$, and if $c = b$, then $a = 0$. Both are excluded by the definition of S^*. Thus all three original elements are distinct. The same tag argument shows that the encoded values are distinct, so allowing repeated values in a zero-sum convention would not alter any oracle answer here.

Conversely, suppose a valid triple satisfies $a + b = c$. When $c = 0$, the initial test finds its opposite pair. Otherwise neither summand is zero: a zero summand would make the other equal to c. Thus $a, b, c \in S^*$. The distinct ranks of a and b differ in some bit $t < L$. Interchange a, b if needed so that their bits are $0, 1$. Then the three corresponding values in (1) sum to zero. The call on this T_t returns YES. □

3 Total time and the exact consequence

Each set T_t can be constructed in $O(n)$ word operations. Its integers have $O(\log n)$ bits, and $\log N = \Theta(\log n)$ because $n \leq N \leq 2n$. Thus the cited algorithm and the new arithmetic use the same logarithmic-word model, with a fixed sufficiently large constant in the word size. The source

algorithm is fixed for $\nu = 6$; sorting, rank assignment, bit tests and the loop over t are also one finite program. No advice or supplied data structure is used.

Including all preprocessing and all $O(\log n)$ source calls, the time is

$$O(n \log n + n^{1.9992} \log n).$$

All logarithms in this time bound may be taken to base two. Put $q = 2500$, $k = \lfloor \log_2 n \rfloor$ and $j = \lfloor k/q \rfloor$. Then $2^{qj} \le n$ and $k + 1 \le q(j + 1)$. The elementary induction $2^j \ge j + 1$ for integers $j \ge 0$ gives

$$\log_2 n < k + 1 \le q(j + 1) \le q2^j \le qn^{1/q}.$$

Consequently the total time is $O(n^{1.9996}) = O(n^{2-1/2500})$, uniformly over S.

Theorem 3. *There is one uniform deterministic classical word-RAM algorithm A, with word size $\Theta(\log n)$, which, for every integer $n \ge 3$ and every set $S \subseteq [-n^3, n^3] \cap \mathbb{Z}$ of cardinality n, correctly decides whether pairwise distinct $a, b, c \in S$ satisfy $a+b = c$. Counting all input-dependent preprocessing, it satisfies*

$$T_A(n) = O(n^{2-1/2500}).$$

Here T_A is the maximum expected step count defined in Section 1; determinism makes the expectation equal to the actual step count. The success probability on every input is 1, and the positive exponent improvement $\varepsilon = 1/2500$ is fixed. The constants do not depend on S or n. The theorem follows from the cited algorithm through Lemma 1 and Proposition 2, and refutes the precise conjecture in the pinned AIM statement. The source of the fixed-power speedup is [1].

References

[1] Josh Alman and Virginia Vassilevska Williams, *Truly Subquadratic 3SUM and Truly Subcubic APSP via Triangles in Sparse Lopsided Graphs*, arXiv:2610.06783v1, October 5, 2026. https://arxiv.org/abs/2610.06783v1. Theorems 2 and 22; Section 3.4.

[2] Steve Brunton, Matthew J. Colbrook, Maarten V. de Hoop, George Stepaniants, Alex Townsend and Rachel Ward, *AIM—Open Applied Problems*, Problem 340, *The modern integer 3SUM conjecture*, revision c9929805cece044705abaaebe1cebddac1ba11a6. https://github.com/MColbrook/AIM/blob/c9929805cece044705abaaebe1cebddac1ba11a6/problems/340-integer-three-sum-hardness.md.

AI use. OpenAI Codex assisted the source search, reduction, checking and drafting of this note under the author's direction. The fast 3SUM algorithm is due entirely to Alman and Vassilevska Williams. Independent mathematical review is requested.